

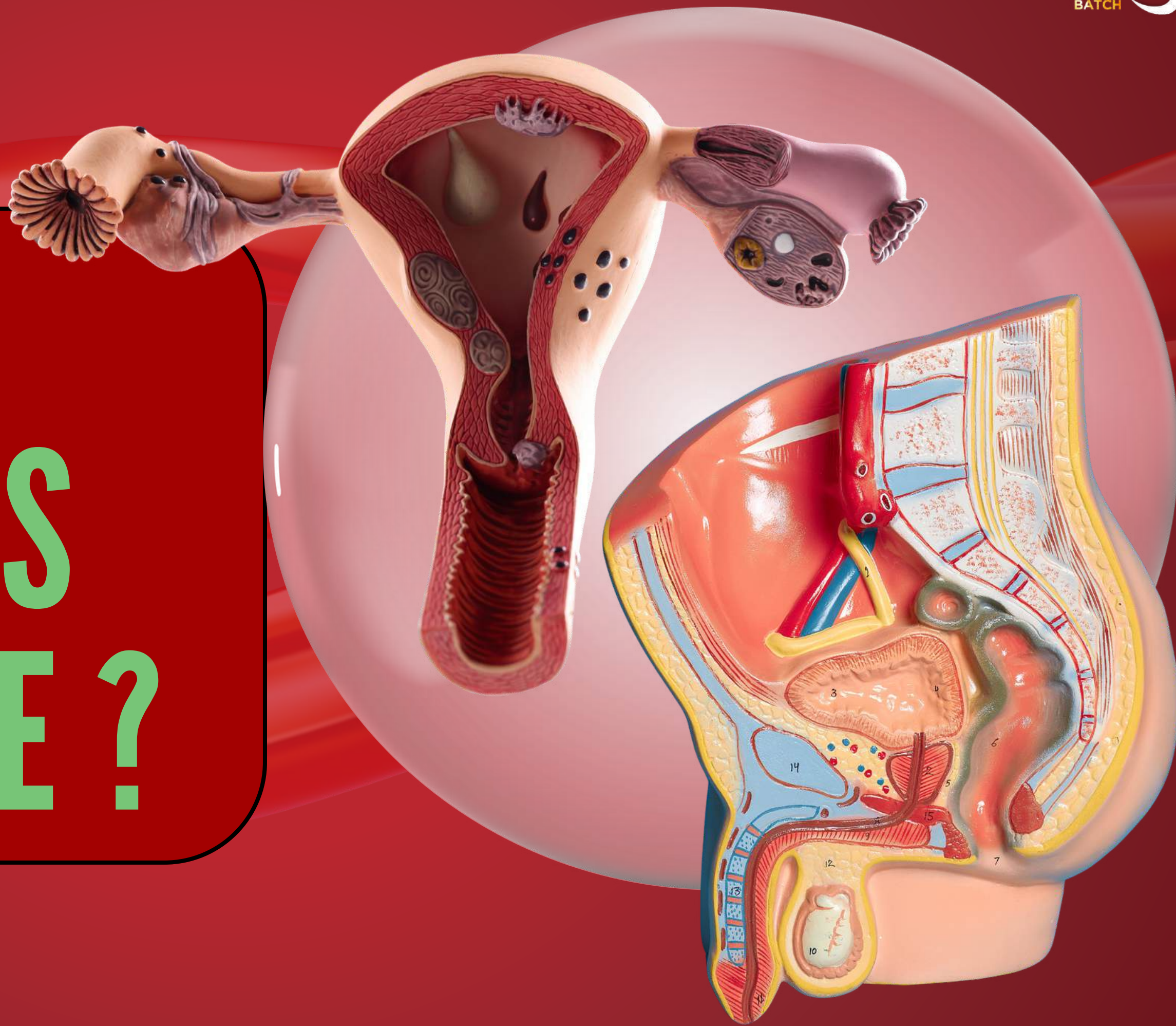
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BATCH

CLASS X - SCIENCE

HOW DO ORGANISMS REPRODUCE ?

PRASHANT KIRAD





Josh Meter

Check yourself

Ques: A student says, Sexual reproduction is simply a more complicated form of asexual reproduction because both ultimately produce a new individual.

Which statement best identifies the fundamental difference?

- a) Asexual reproduction always occurs in unicellular Organisms.**
- b) Sexual reproduction involves the participation of two individuals in producing a new generation.**
- c) Asexual reproduction never involves DNA copying.**
- d) Sexual reproduction cannot occur in multicellular organisms.**

Check yourself

Ques: The reason for the greater similarities among offsprings produced by asexual reproduction is

- 1. Asexual reproduction involves only one parent**
- 2. Asexual reproduction involves two parents**
- 3. Asexual reproduction involves gametes**
- 4. Asexual reproduction does not involve gametes**

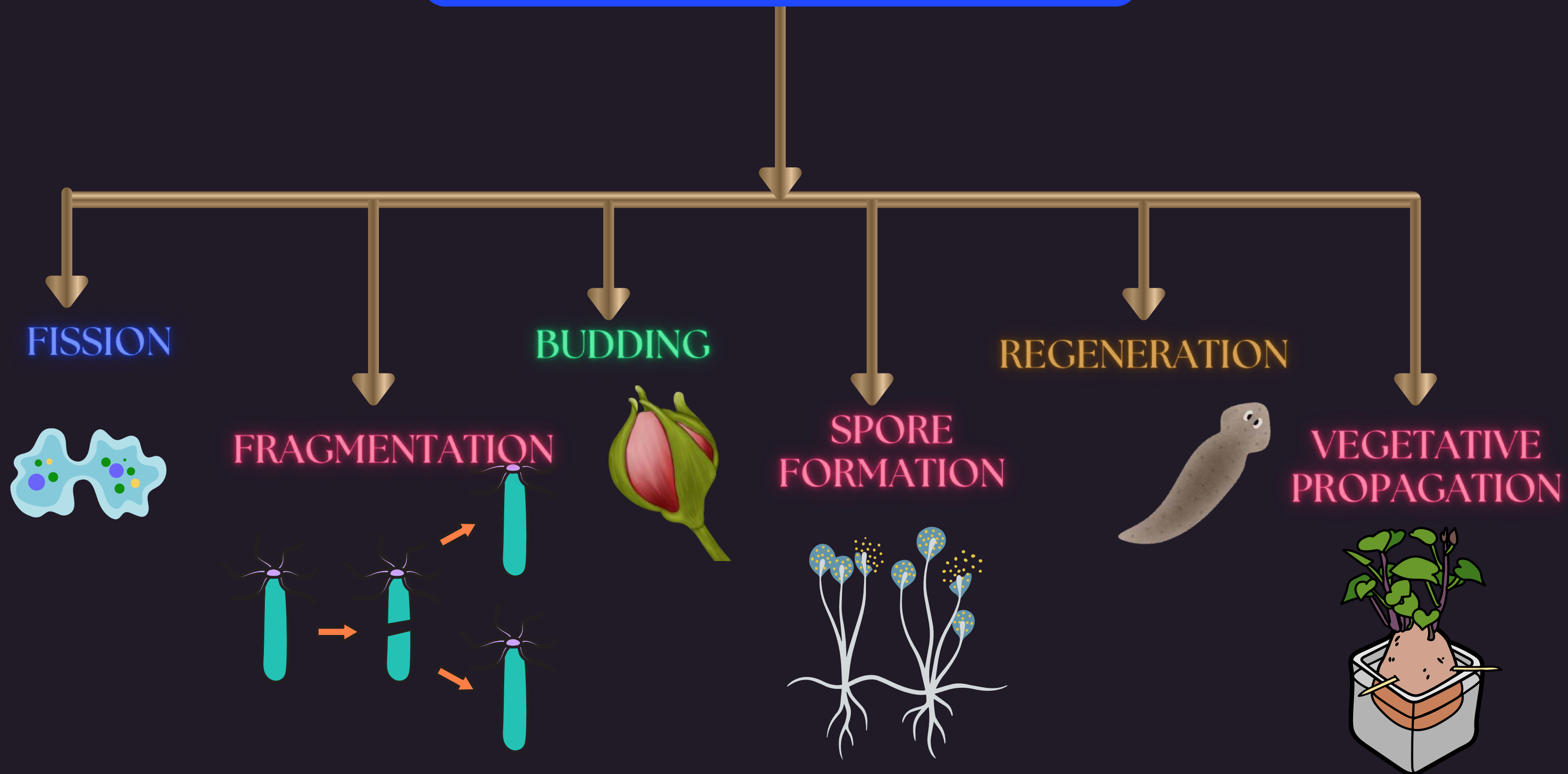
- (a) 1 and 2**
(b) 1 and 3
(c) 2 and 4
(d) 1 and 4

Check yourself

Ques: A species contains two groups of individuals. Group X reproduces using a process involving one individual, whereas Group Y uses a process involving two individuals.

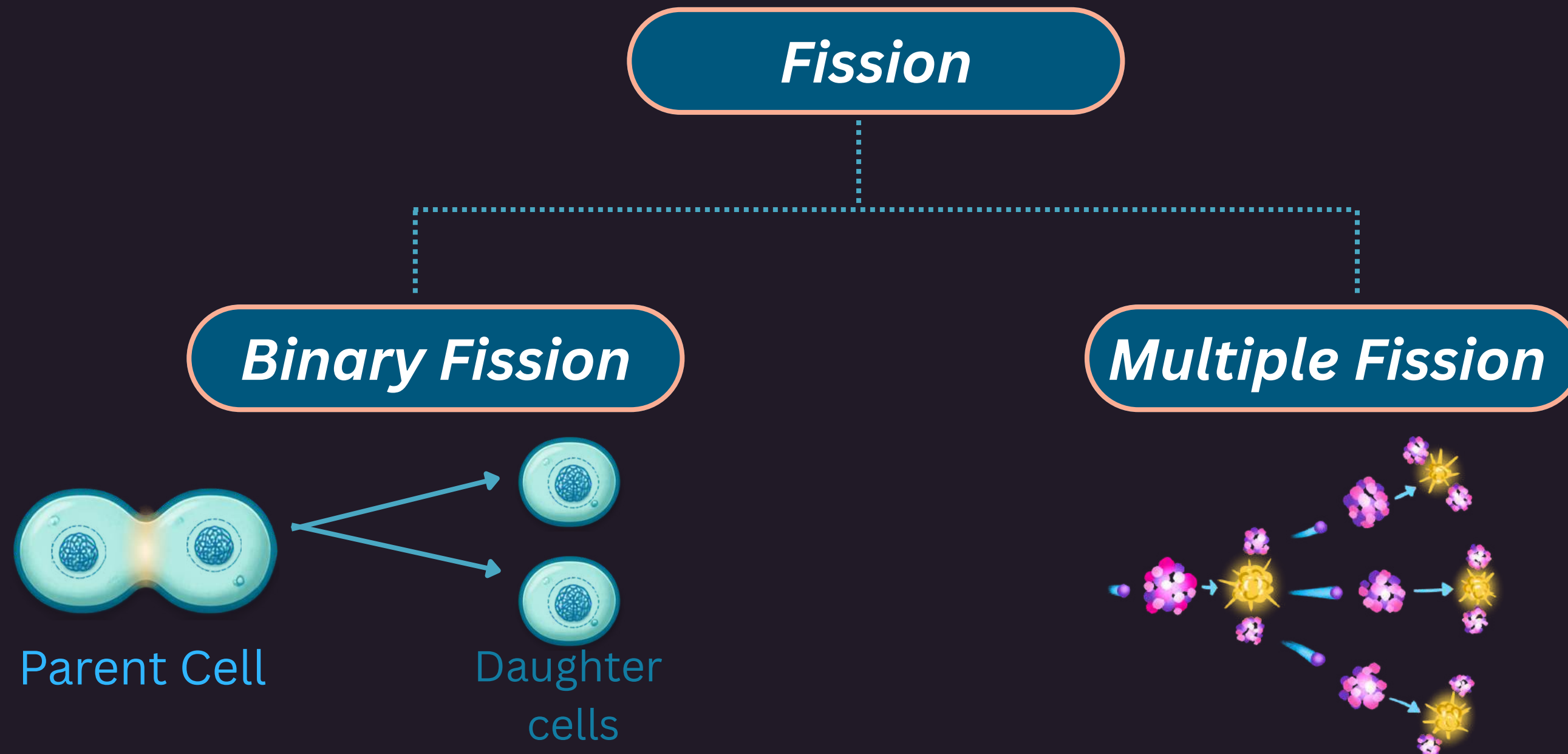
- a) Identify the two broad reproductive modes.**
- b) Which mode is expected to generate newer combination of variation more effectively?**
- c) Explain why these variations can be significant at the population level?**

ASEXUAL REPRODUCTION



Fission

The process of reproduction in unicellular organisms in which a parent cell divides into two or more new individuals.



Binary Fission

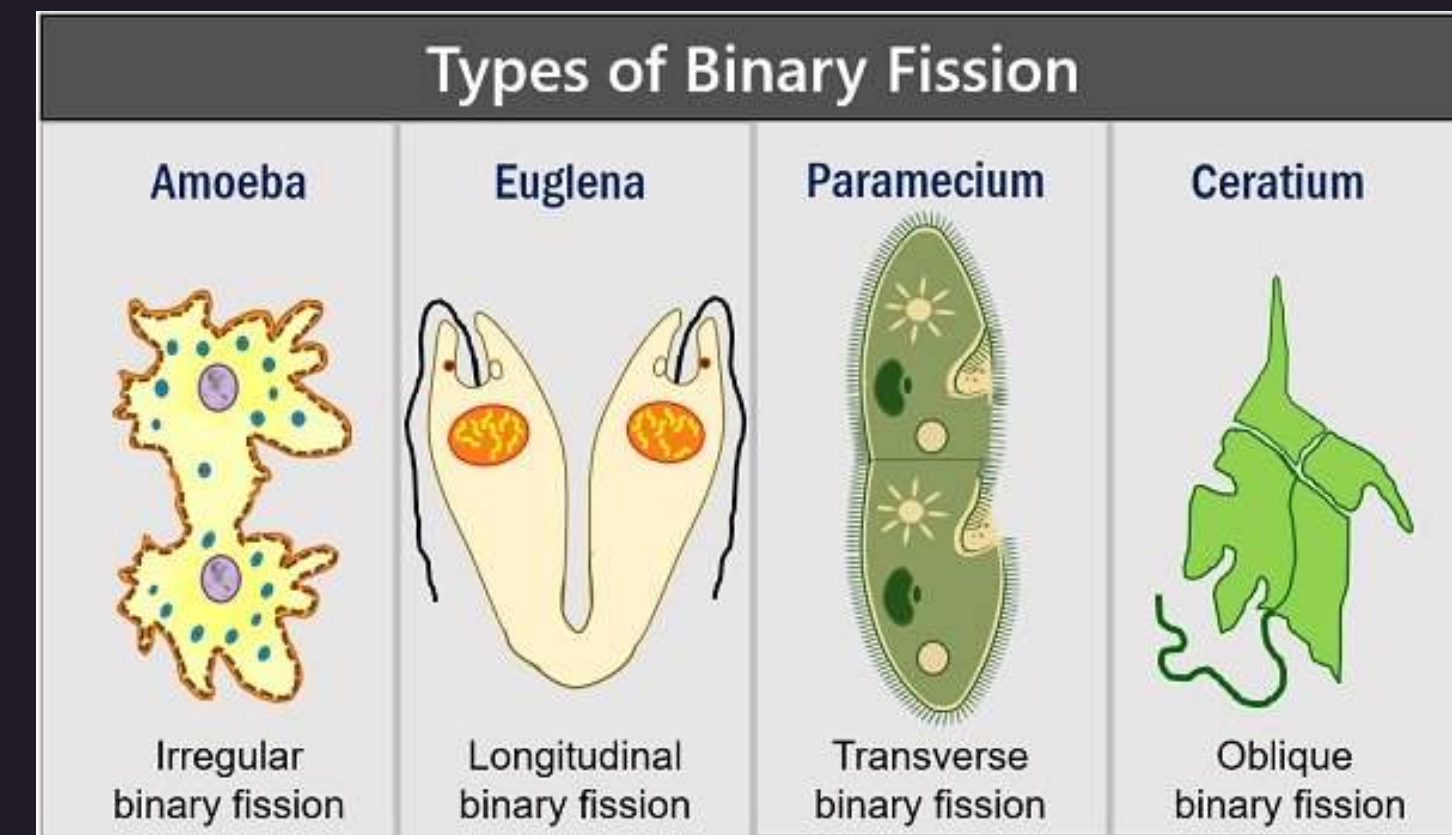
Fission → The process of reproduction in unicellular organisms in which a parent cell divides into two or more new individuals.

Binary fission is a type of asexual reproduction in which a unicellular parent organism divides into two daughter cells.

- The nucleus divides first, followed by the **division of cytoplasm** (cytokinesis).
- Each daughter cell receives **a copy of DNA** and develops into a complete new organism.
- **Examples: Amoeba, Paramecium, Euglena.**



Type of Binary Fission	Plane of Division	Example
Irregular	Any plane	<i>Amoeba</i>
Longitudinal	Along the length	<i>Euglena</i>
Transverse	Across the width	<i>Paramecium</i>
Oblique	At an angle	<i>Some algae, Dinoflagellates</i>



Amoeba – Unicellular organism

- Splitting of the cell can take place in any plane
- Parent cell divides into two daughter cells



Leishmania – Unicellular organism

- Has a whip-like structure at one end of the cell
- Binary fission occurs in a definite orientation in relation to the structure
- Causes **Kala-azar** disease.

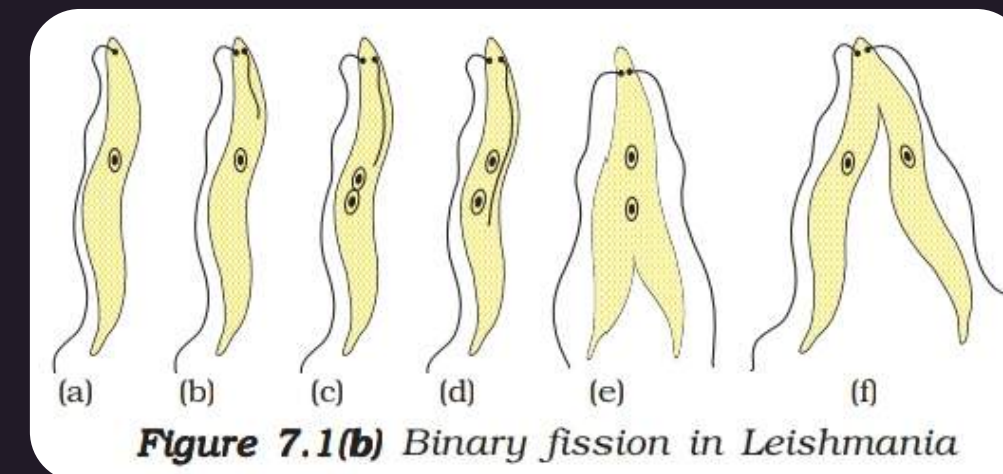
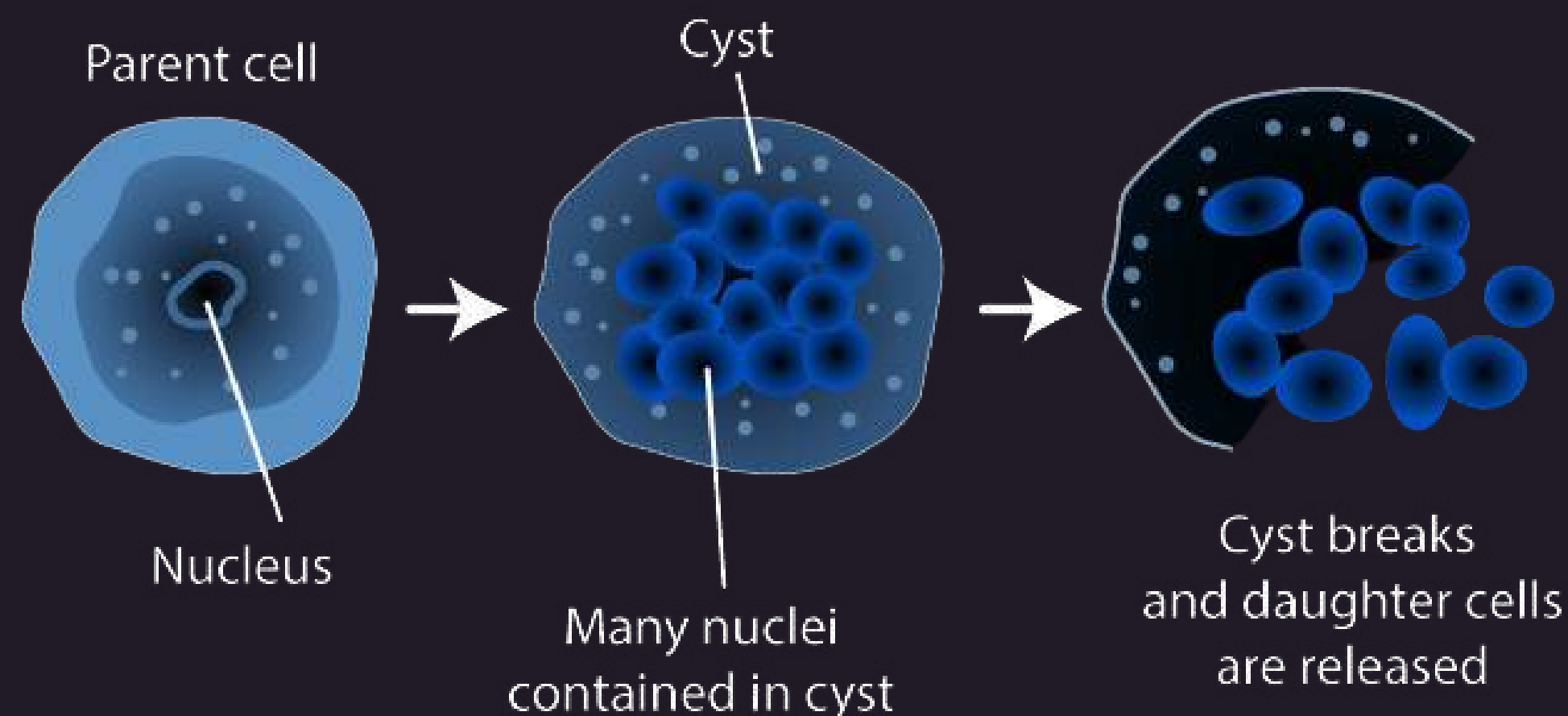


Figure 7.1(b) Binary fission in *Leishmania*

MULTIPLE FISSION: IS AN ASEYUAL MODE OF REPRODUCTION IN WHICH A SINGLE PARENT CELL DIVIDES TO FORM MANY DAUGHTER CELLS.

Plasmodium – Unicellular organism

- First, the nucleus divides repeatedly to produce several nuclei.
- Then, the cytoplasm divides, forming many daughter cells inside the parent cell.
- Under unfavourable conditions, a cyst wall forms around the parent cell for protection.
- When conditions become favourable, the cyst wall breaks, releasing all the daughter cells.
- Example: Plasmodium (malarial parasite).



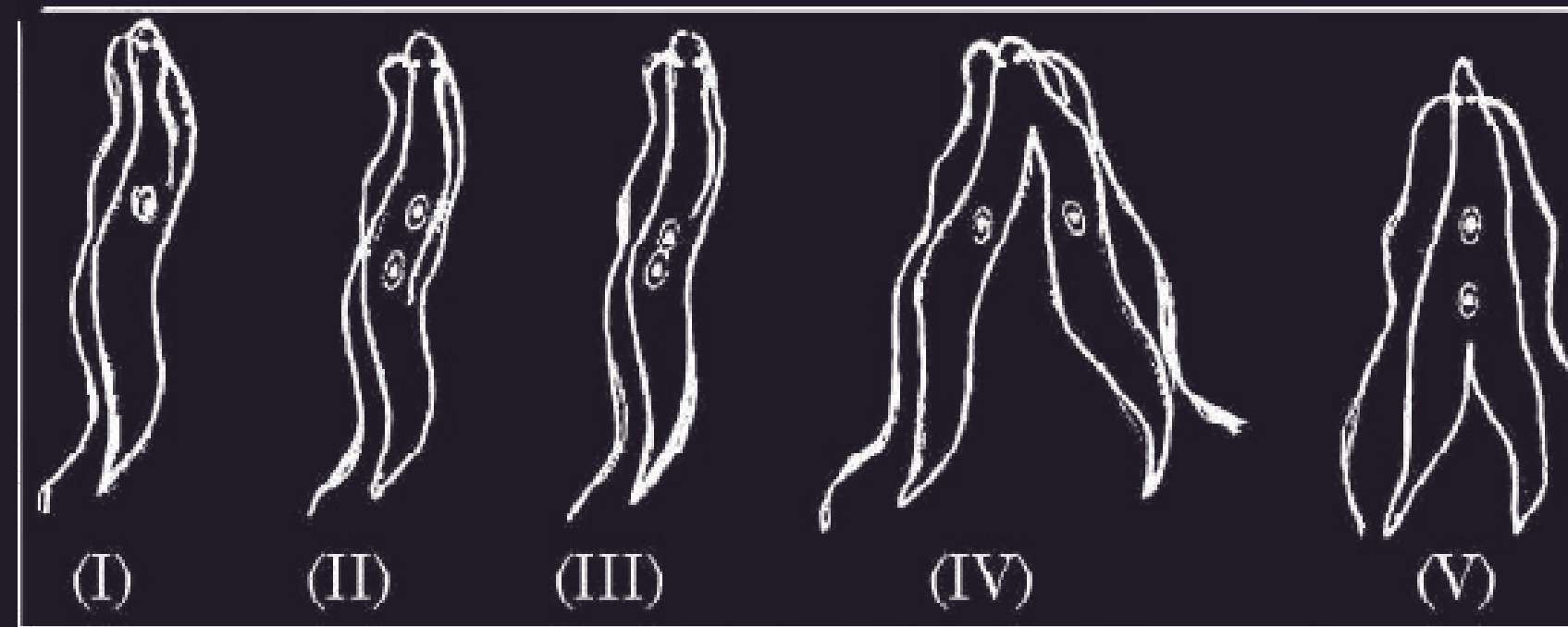
Check yourself

Ques: How does Binary Fission differ from multiple fission? [NCERT]

Binary Fission	Multiple Fission
Parent cell divides into two daughter cells .	Parent cell divides into many daughter cells at the same time.
Division can occur in any plane (e.g., Amoeba) or in a definite orientation (e.g., Leishmania).	Nucleus divides repeatedly to form many nuclei; later the cytoplasm divides, enclosing each nucleus (e.g., Plasmodium).
Simple and relatively fast process.	More complex and comparatively slow process.
Example: Amoeba, Leishmania, Paramecium.	Example: Plasmodium (malarial parasite).

Check yourself

Q. Choose the correct sequence of stages of binary fission in Leishmania.



(a) I, II, III, IV, V

(b) I, III, II, V, IV

(c) I, III, V, II, IV

(d) I, II, III, V, IV

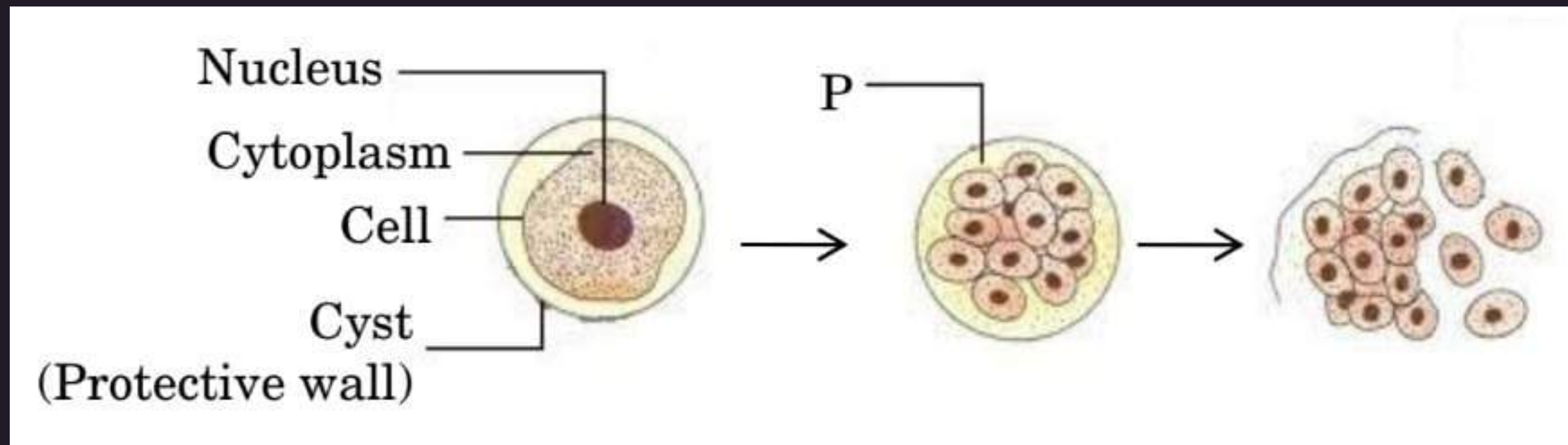
Check yourself

Q. Name the causative organism of Kala-azar and state its mode of asexual reproduction.

Answer: Leishmania; binary fission.

Check yourself

Ques: Observe the given diagram and answer the questions that follow:[5 Marks]



- (i) Give the name and type of reproduction shown in the above diagram.
- (ii) Comment on the processes taking place in stage P.
- (iii) Name an organism which reproduces by the above mentioned method.

2015, 2016, 2017, 2019, 2024

Q. How do Leishmania and Plasmodium reproduce, and mention the difference in their modes of reproduction?

Leishmania reproduces by binary fission.
 The parent cell divides into two daughter cells in a definite orientation – along a particular axis, related to its whip-like flagellum.
 (Disease caused: Kala-azar)

Plasmodium reproduces by multiple fission.
 The parent cell’s nucleus divides several times to form many daughter nuclei, each developing into a new organism after the cell bursts.
 (Disease caused: Malaria)

Organism	Mode of Reproduction	Type of Fission	Number of Offspring Produced
<i>Leishmania</i>	Asexual	Binary Fission	Two daughter cells
<i>Plasmodium</i>	Asexual	Multiple Fission	Many daughter cells

Regeneration

The ability of fully differentiated organisms to give rise to new individuals from their body parts when cut or broken into pieces.

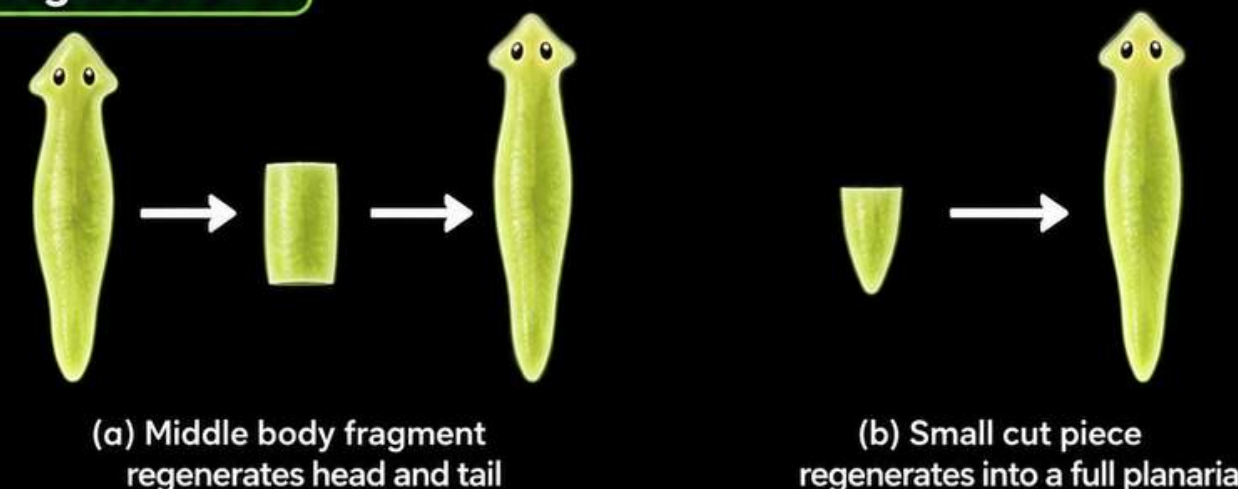
Example - Hydra and Planaria can be cut into many pieces, and each piece grows into a complete organism.

Process:

- Carried out by specialised cells.
- These cells proliferate (multiply) and form a mass of cells.
- This mass undergoes differentiation → develops into various cell types and tissues.
- The changes occur in an organised sequence called development.

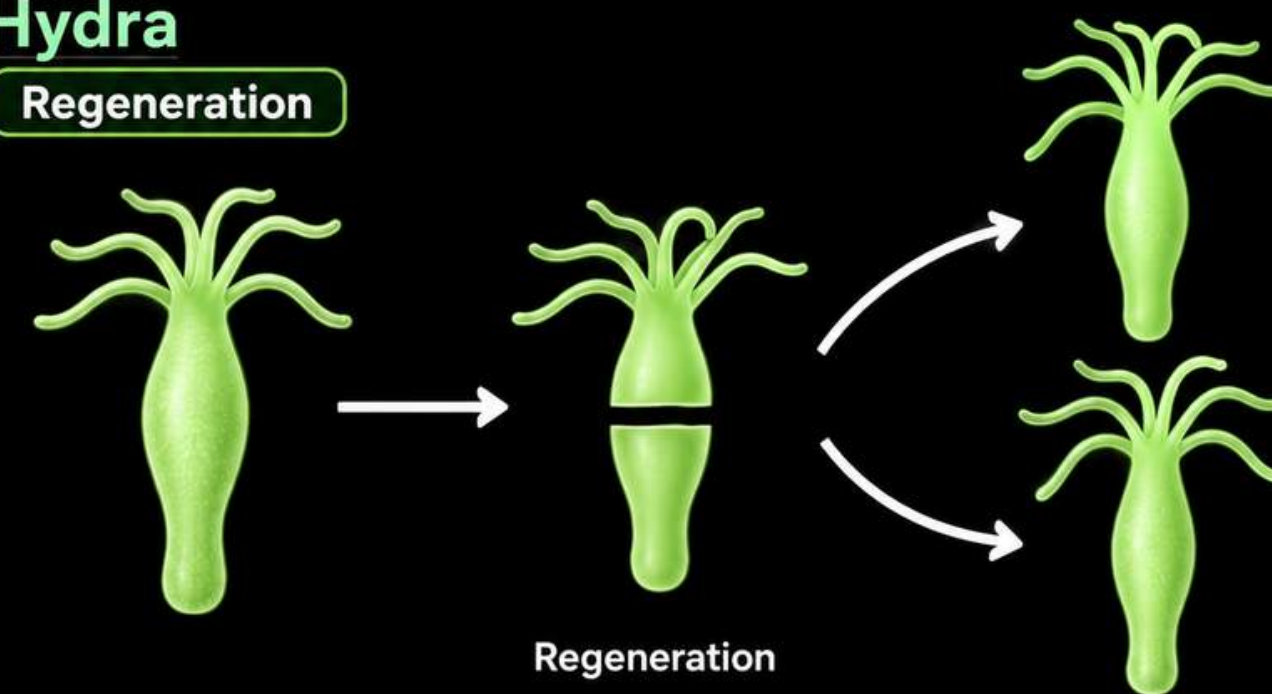
Planaria

Regeneration



Hydra

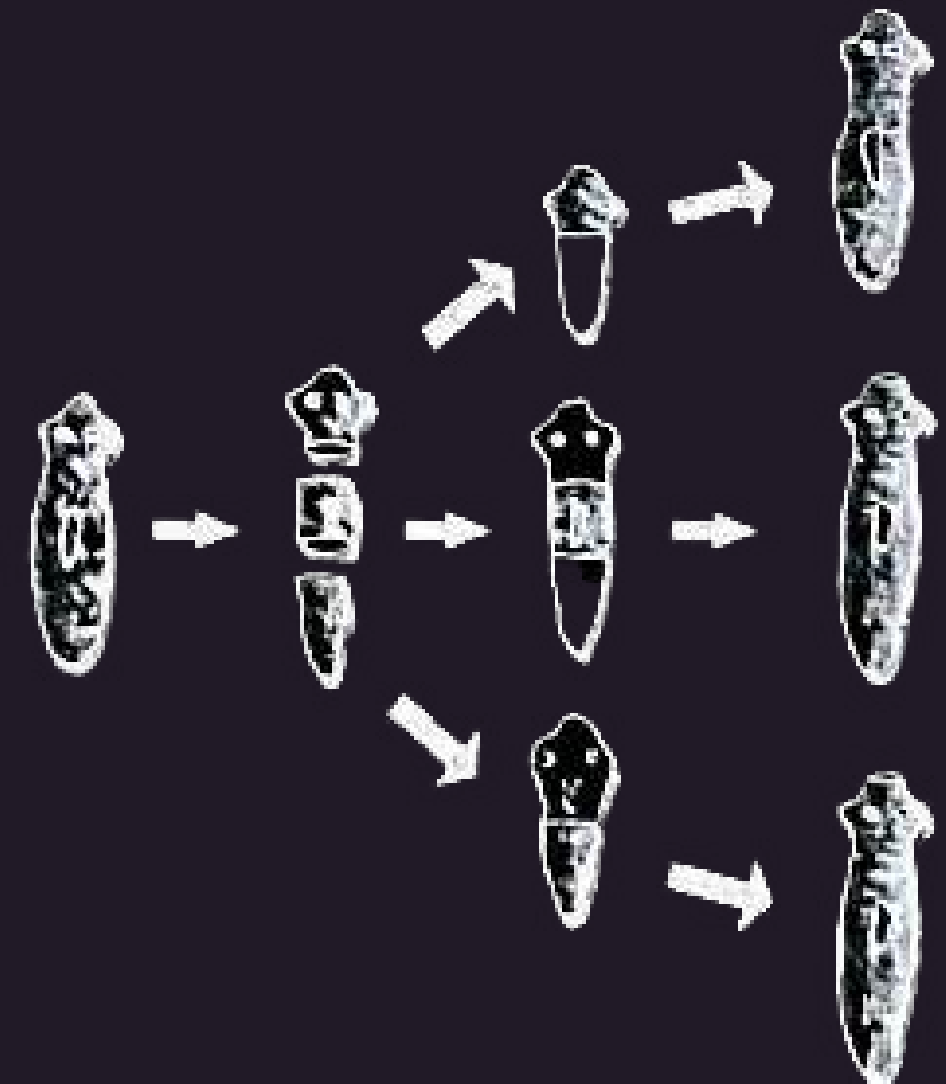
Regeneration



Check yourself

Q. (a) Name the process shown below and define it:

(b) Name the type of cells present in the organisms which exhibit this process.



Check yourself

Q. A student claims, “If regeneration can produce complete organisms from body parts, then every multicellular organism should be capable of reproducing by regeneration.” Evaluate this statement.

Ans. The statement is incorrect.

- Regeneration occurs only in some relatively simple multicellular organisms, such as Planaria and Hydra.
- In these organisms, specialised cells can divide, form a mass of cells, and then differentiate into different tissues and organs.
- In complex multicellular organisms, cells are highly organised into specialised tissues and organs, so a separated body part usually cannot develop into a complete organism.

Check yourself

2015, 2017, 2019; Sample Paper 2022

Q. How is the process of regeneration different from reproduction? Explain with example.

Regeneration: It is the ability of an organism to regrow lost body parts. In some simple organisms, a body fragment can even develop into a complete organism through cell division and differentiation.

Example: Each body fragment of Planaria can regenerate into a complete Planaria.

Reproduction: It is the biological process by which organisms produce new individuals of their own species as a normal part of their life cycle.

Example: Hydra reproduces by budding.

Thus, regeneration is not the same as reproduction because organisms do not normally depend on being cut into pieces to reproduce.

Check yourself

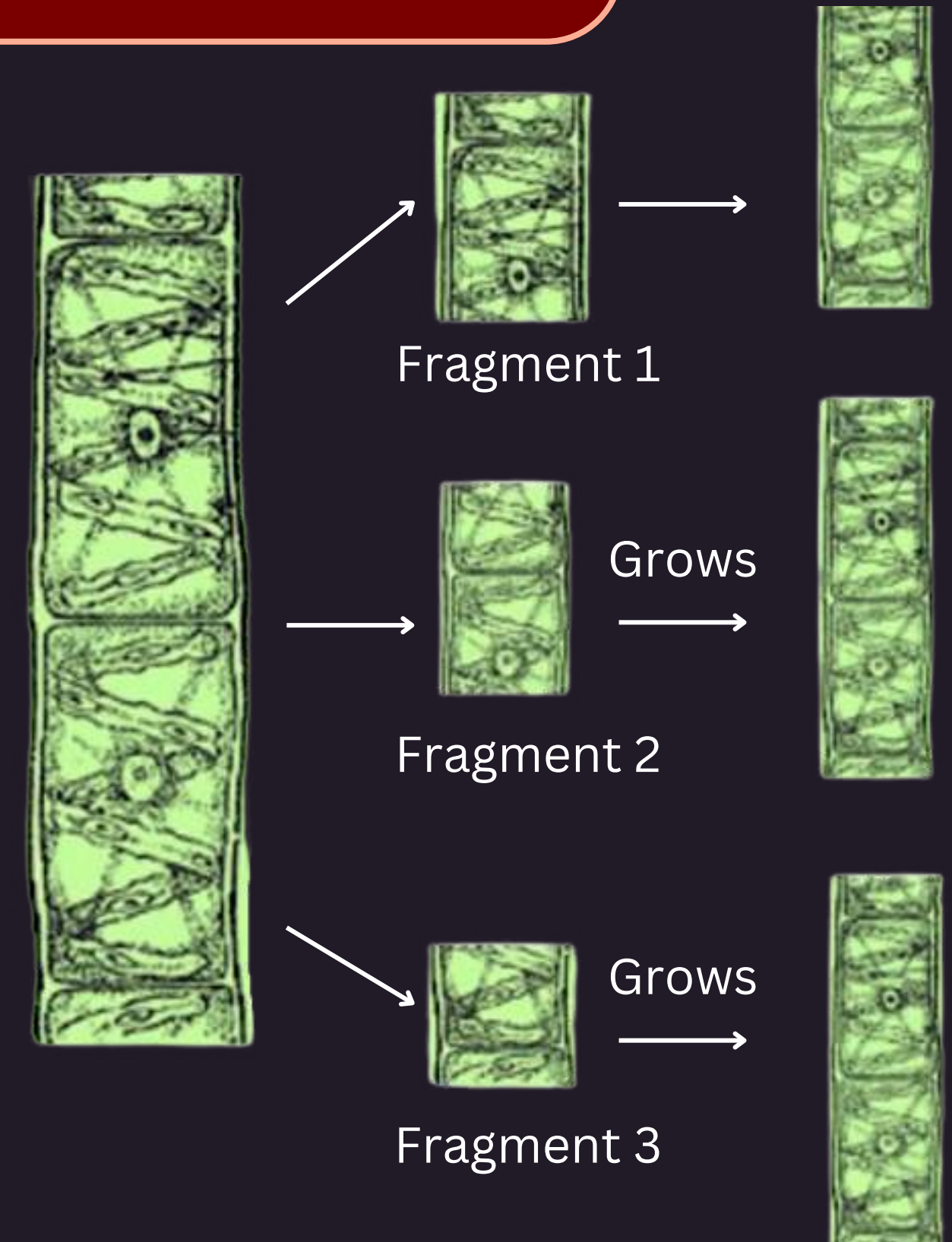
CBSE 2015, 2020, 2022, 2024

Q. Why is regeneration possible only in simple organisms like Hydra and Planaria but not in complex organisms?

- **Regeneration occurs only in simple organisms because they have simple body design and less cell specialization. Their cells can divide and form all body parts.**
- **In complex organisms, cells are highly specialized, so they can only repair tissues, not regenerate the whole body.**

Fragmentation

- *It is a type of asexual reproduction in which an organism breaks up into two or more fragments, and each fragment grows into a new individual. Example: **Spirogyra***
- *Occurs when the organism matures.*
- *Happens in favourable conditions – enough water, nutrients, and sunlight.*
- *In **complex multicellular organisms**, this type of reproduction is not possible because their cells are organised into tissues and organs.*



Check yourself

Ques: A filamentous Organism has a relatively simple body organisation. On maturation, the filament breaks into several pieces, each piece develops into a new individual. Which featyr of the Organism makes the reproductive strategy possible?

- a) It's body consists of specialised organs arranged in complex system**
- b) It's relatively simple organisation allows the fragments to develop into individuals.**
- c) It produces gametes that fuse externally.**
- d) It forms seeds before fragmentation**

Check yourself

2015, 2018, 2020, CBSE Sample Paper: 2023, 2024

Q. Which of the following statements is correct about Spirogyra and Planaria?

- a) Spirogyra reproduces by regeneration and Planaria by fragmentation.**
- b) Spirogyra reproduces by fragmentation and Planaria by regeneration.**
- c) Both Spirogyra and Planaria reproduce by fission.**
- d) Both Spirogyra and Planaria reproduce by budding.**

Check yourself

2013, 2017, 2020

Q. The process of breaking up of an organism into two or more small pieces, each of which grows into a new individual, is called:

- (a) Budding
- (b) Fragmentation
- (c) Regeneration
- (d) Spore formation

Check yourself

Q. Water in a pond appears dark green and contains filamentous structures. Name these structures and the method by which they reproduce. Explain the process.

Ans. The filamentous structures are Spirogyra.

They reproduce asexually by fragmentation.

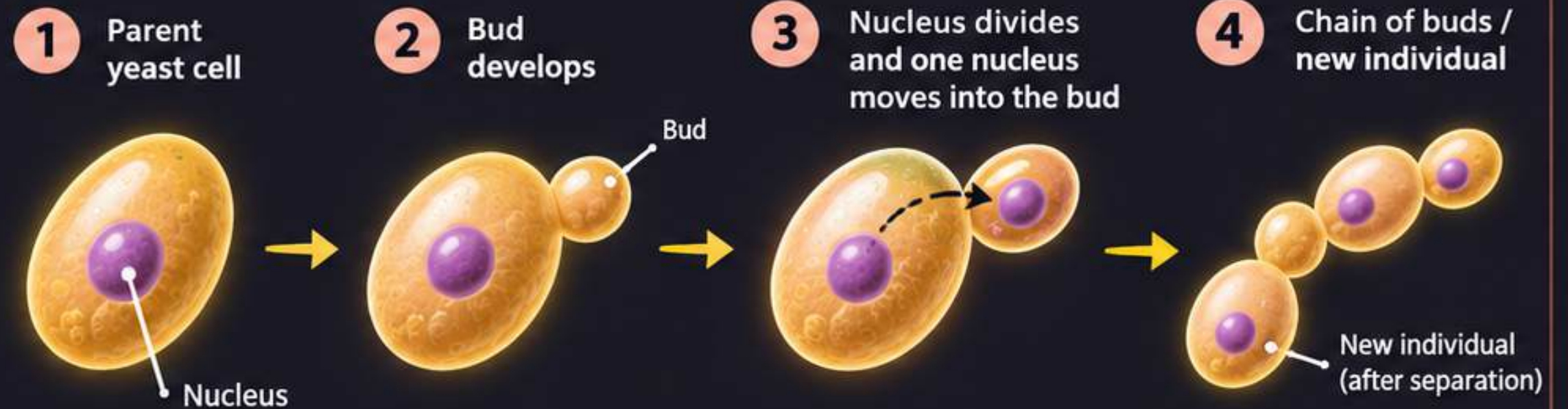
Process: The mature Spirogyra filament breaks into two or more fragments. Each fragment grows independently and develops into a new individual under favourable conditions.

Budding

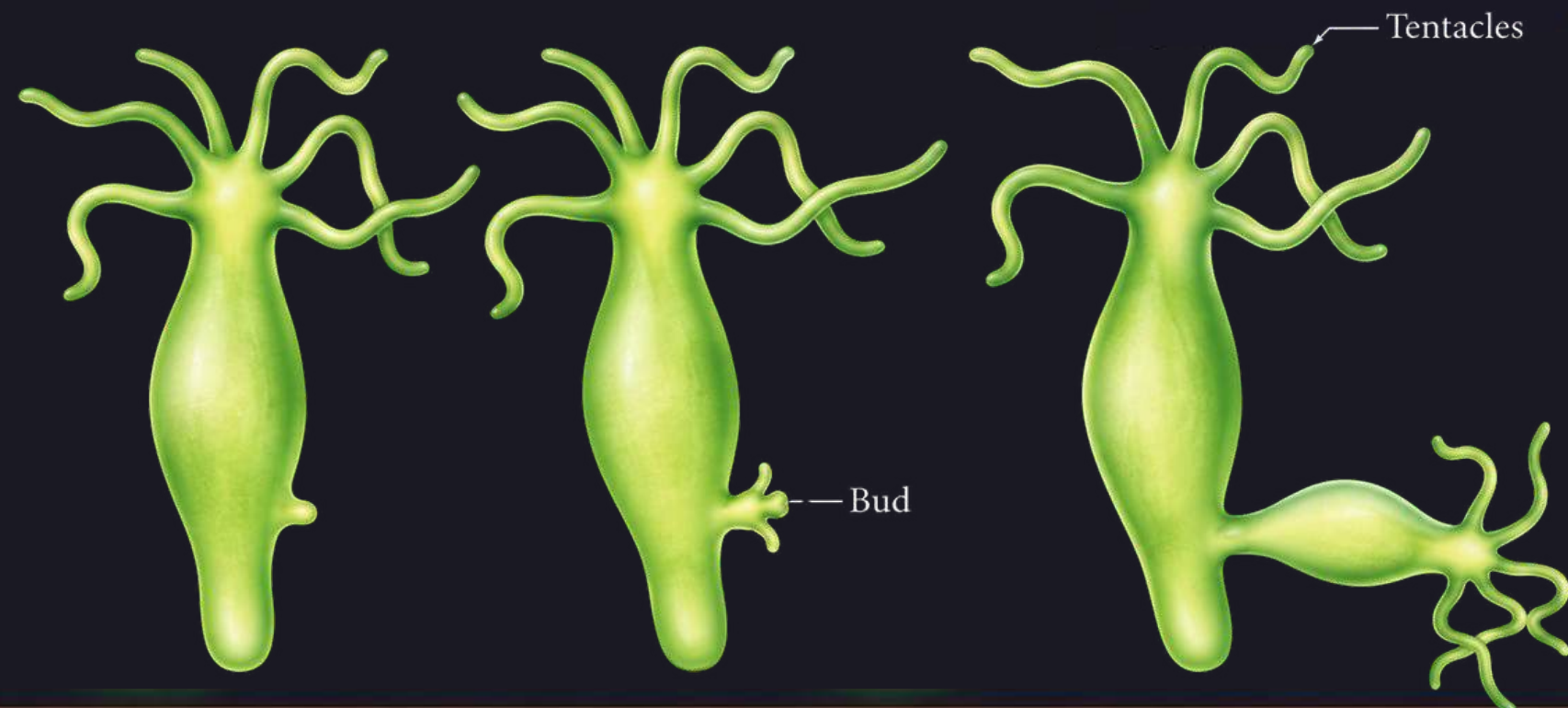
A type of **asexual reproduction** in which a small outgrowth (**bud**) develops on the parent body, grows by **repeated cell division**, and finally detaches to form a **new individual**.

Examples: Hydra & Yeast

Budding in Yeast



Budding in Hydra

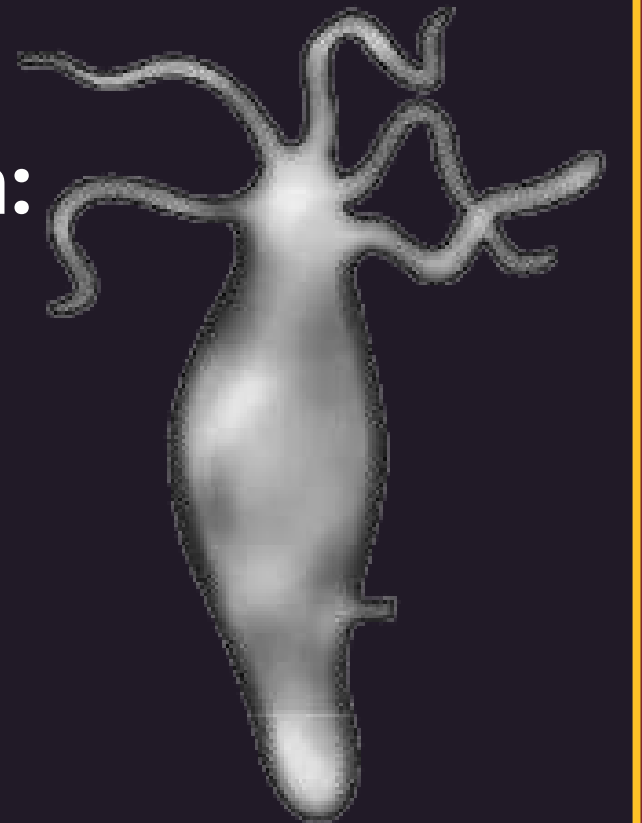


Check yourself

2024

Q. Identify the mode of asexual reproduction in the following organism:

- (a) Fragmentation
- (b) Multiple fission
- (c) Budding
- (d) Binary fission



Check yourself

Q. Assertion (A): Hydra reproduces asexually by budding.

Reason (R): A small outgrowth develops on the parent body due to repeated cell division and later separates as a new individual.

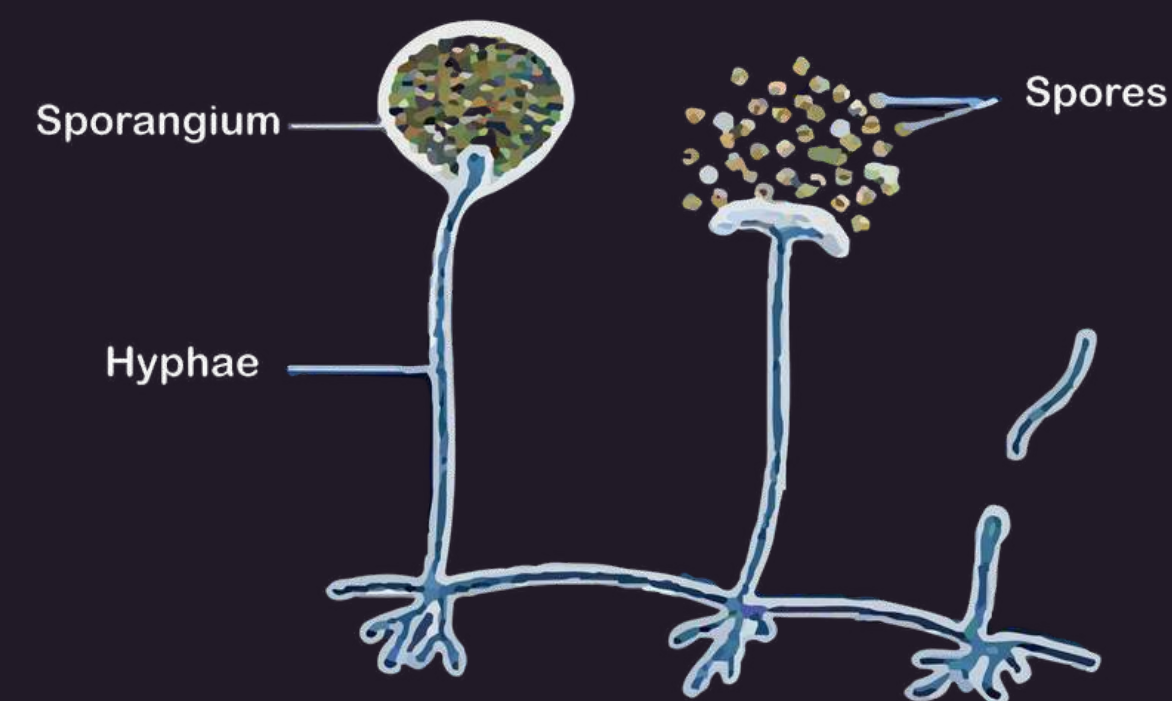
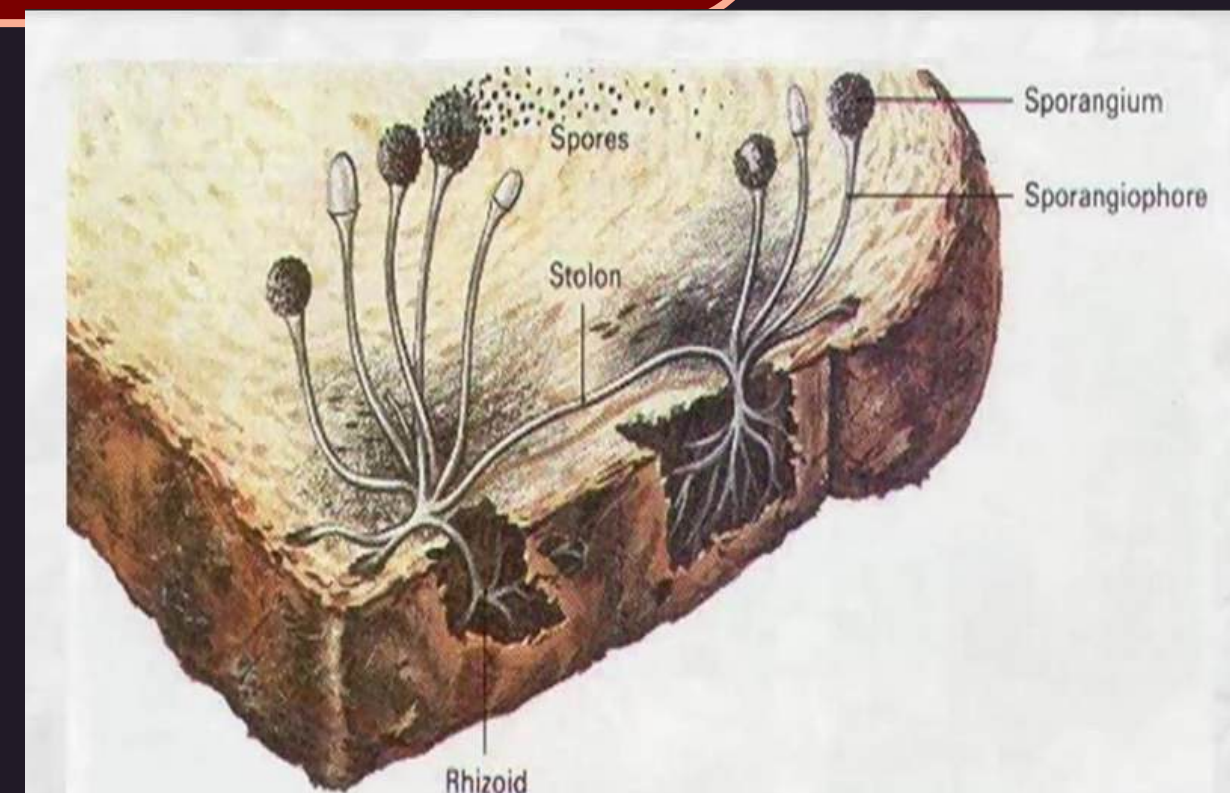
- (a)** Both A and R are true, and R is the correct explanation of A.
- (b)** Both A and R are true, but R is not the correct explanation of A.
- (c)** A is true, but R is false.
- (d)** A is false, but R is true.

Spore Formation

A type of asexual reproduction in which new individuals are formed from tiny spores produced inside special reproductive structures called sporangia.

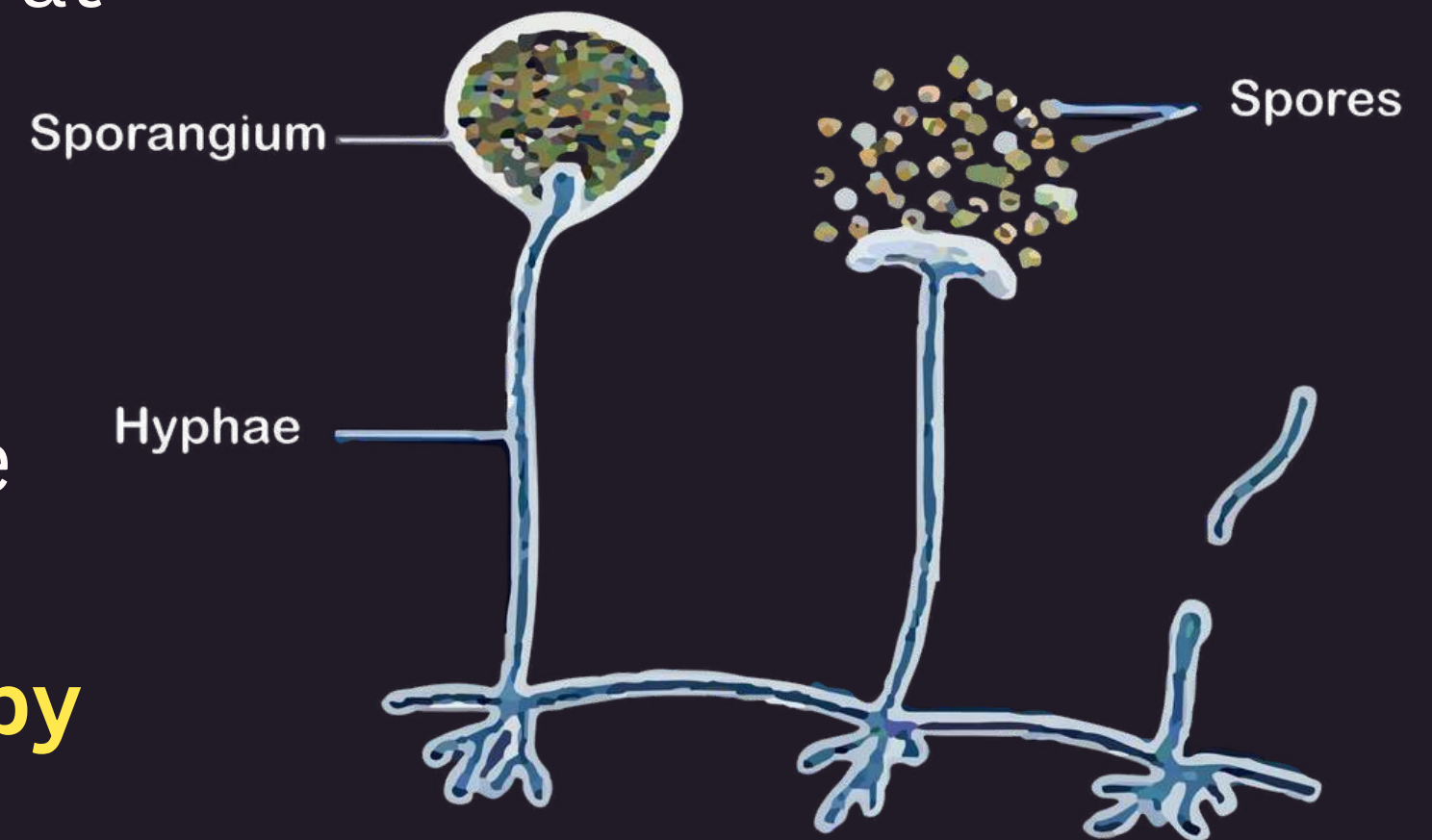
In Rhizopus (bread mould):

- The thread-like structures on bread are hyphae (non-reproductive).
- The tiny blob-on-a-stick structures are sporangia.
- Sporangia contain spores which can develop into new individuals.
- Spores have thick protective walls that help them survive unfavourable conditions.
- When spores land on a moist surface, they germinate and grow into a new Rhizopus.



Advantages of Spore Formation:

- **Large number of spores** are produced at a time, increasing the chances of survival.
- **Spores have thick protective walls**, which help them survive unfavourable conditions.
- Spores are **light and easily dispersed by air to new places**.
- When they **reach favourable conditions**, they germinate and develop into new individuals.



Check yourself

Ques: How will an organism be benefitted if it reproduces through spores?
[NCERT]

Ans. Spore formation is advantageous because spores have thick protective walls that help them survive unfavourable conditions. They remain dormant and germinate when favourable conditions return, forming new individuals.

Check yourself

Q. Name the reproductive and non-reproductive parts of bread mould (Rhizopus).

Check yourself

Q. Which of the following conditions are responsible for the rapid spread of bread mould?

- 1. Production of a large number of spores**
- 2. Presence of moisture and nutrients**
- 3. Low temperature**
- 4. Presence of hyphae**

2015, 2016, 2017, 2018, 2020, 2023, 2024

Q.Explain reproduction by spores in Rhizopus with the help of a labelled diagram. Why does Rhizopus not multiply on a dry slice of bread?

Rhizopus (bread mould) reproduces by spore formation.

It has thread-like structures called hyphae, which form a cotton-like mass called mycelium.

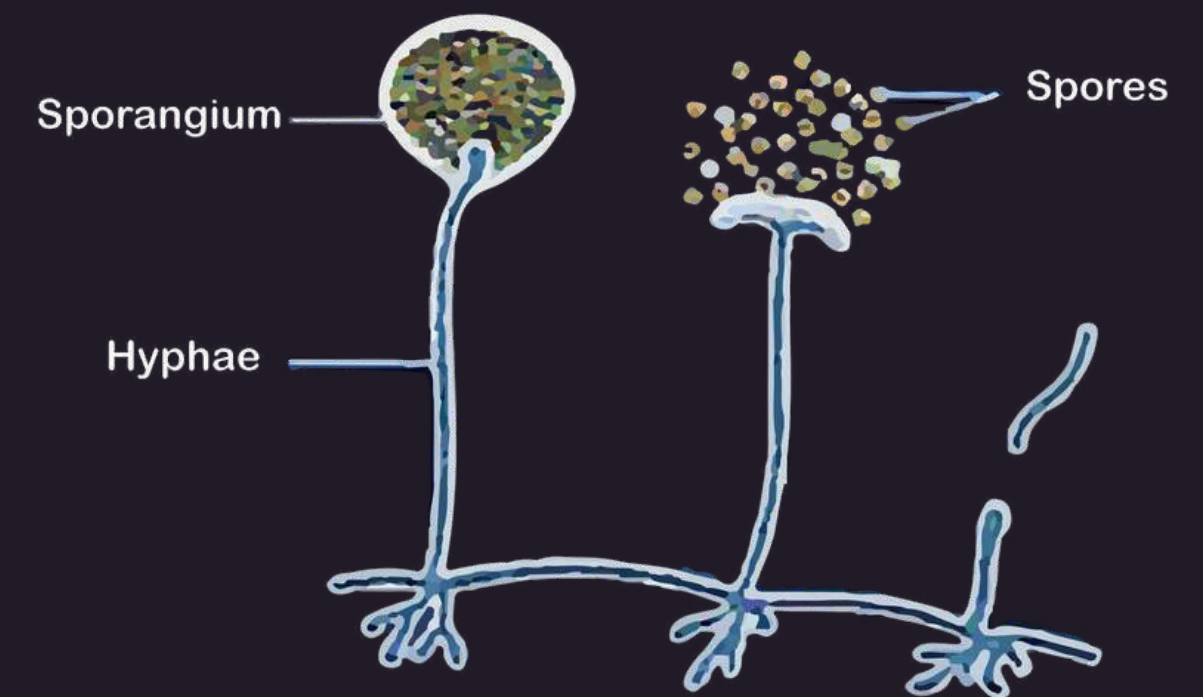
At the tip of upright hyphae, sporangia develop that contain numerous spores.

When the sporangium bursts, spores are released into the air.

Under favourable conditions (moisture, warmth, and nutrients), each spore germinates to form a new fungus.

Reason:

Rhizopus does not multiply on dry bread because moisture is essential for spore germination. Dry conditions prevent the spores from growing.



Check yourself

Q. Select the option having correct matching of the organism given in Column I with the mode of reproduction in Column II:

- (a) P-4, Q-2, R-1, S-3
- (b) P-3, Q-4, R-5, S-2
- (c) P-3, Q-4, R-1, S-2
- (d) P-4, Q-3, R-2, S-1

Column I	Column II
P. Leishmania	1. Regeneration
Q. Spirogyra	2. Multiple Fission
R. Planaria	3. Binary Fission
S. Plasmodium	4. Fragmentation
	5. Budding

Vegetative Propagation

A type of asexual reproduction in which roots, stems, or leaves of a plant develop into a new plant under suitable conditions.

Parts of a Plant



Reproductive part

This part of the plant (flower) which is involved in the process of sexual reproduction.



Vegetative part

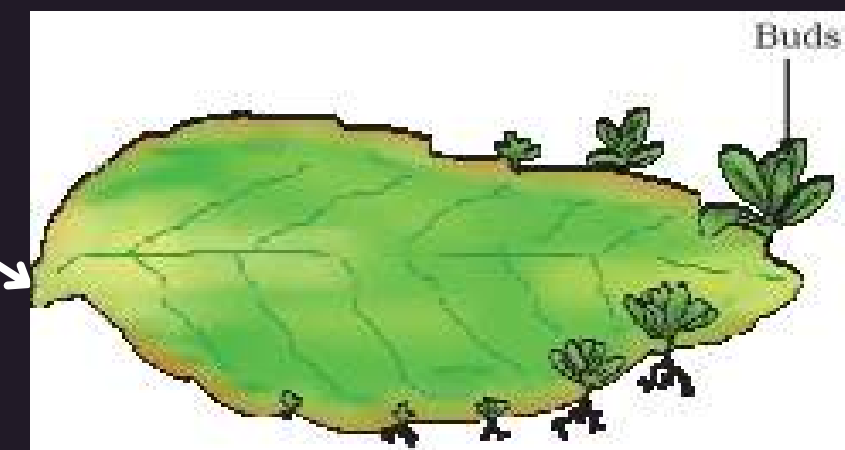
These parts of the plant (stem, leaf, root) are not involved in the process of sexual reproduction.



Natural Vegetation

Plants can reproduce naturally without human interference through their roots, stems, or leaves. New plants develop from these parts of the parent plant.

Mode of Vegetative Propagation	Organism / Example
By Underground Stem (Tubers)	<i>Potato</i>
By Roots (Tuberous Roots)	<i>Sweet Potato, Dahlia</i>
By Leaves (Leaf Buds)	<i>Bryophyllum, Kalanchoe</i>



Artificial Vegetative Propagation:

It is the man-made, human-controlled method of reproduction in which new plants are produced from vegetative parts (stem, root, leaf) of the parent plant using special techniques.

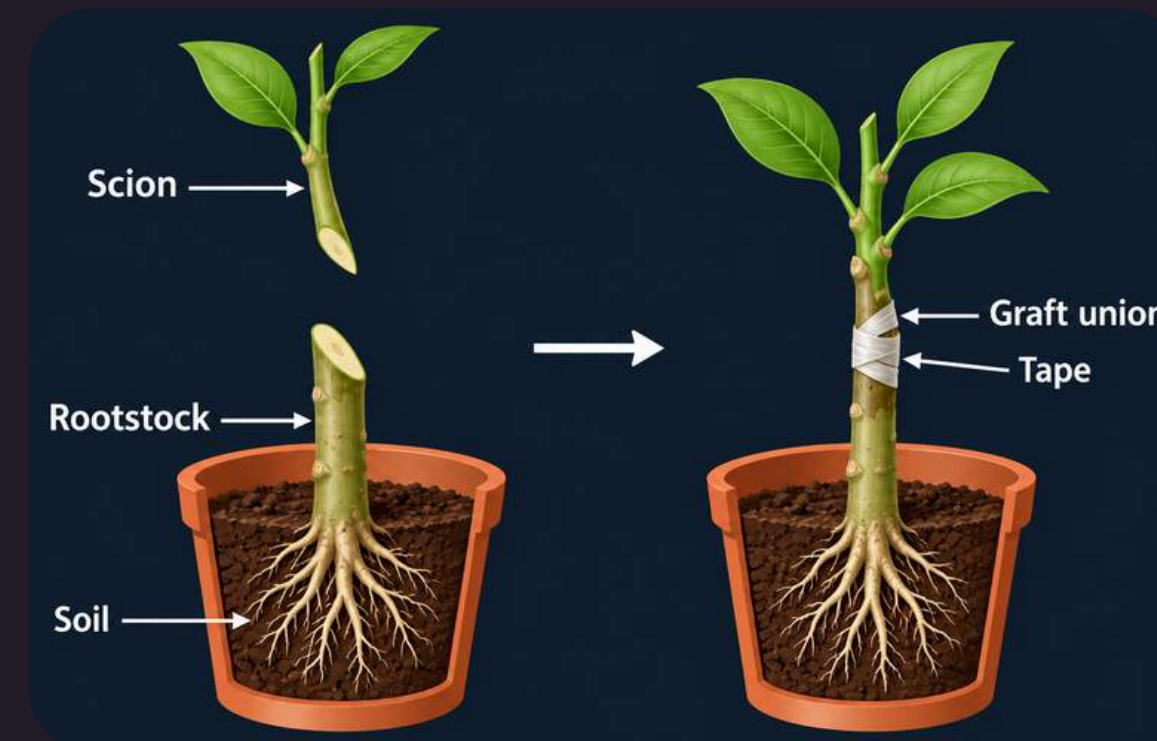
Cutting:

- A piece of stem or leaf is cut and planted in soil to grow a new plant. Examples: Rose, Hibiscus, Money plant, Sugarcane.



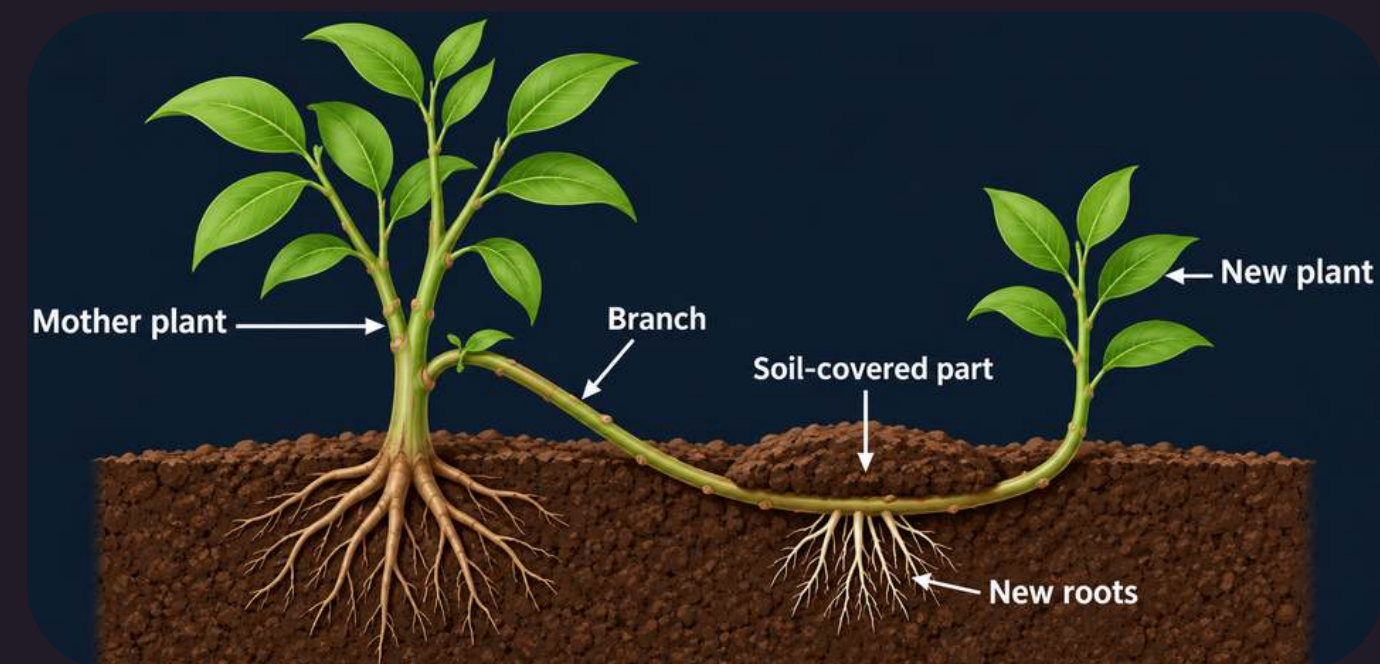
Grafting:

- In grafting, the stem/branch of one plant called the **scion** is joined to the rooted plant called the **stock**. They grow together as a single plant. Examples: Mango, Apple, Rose, Grapes.



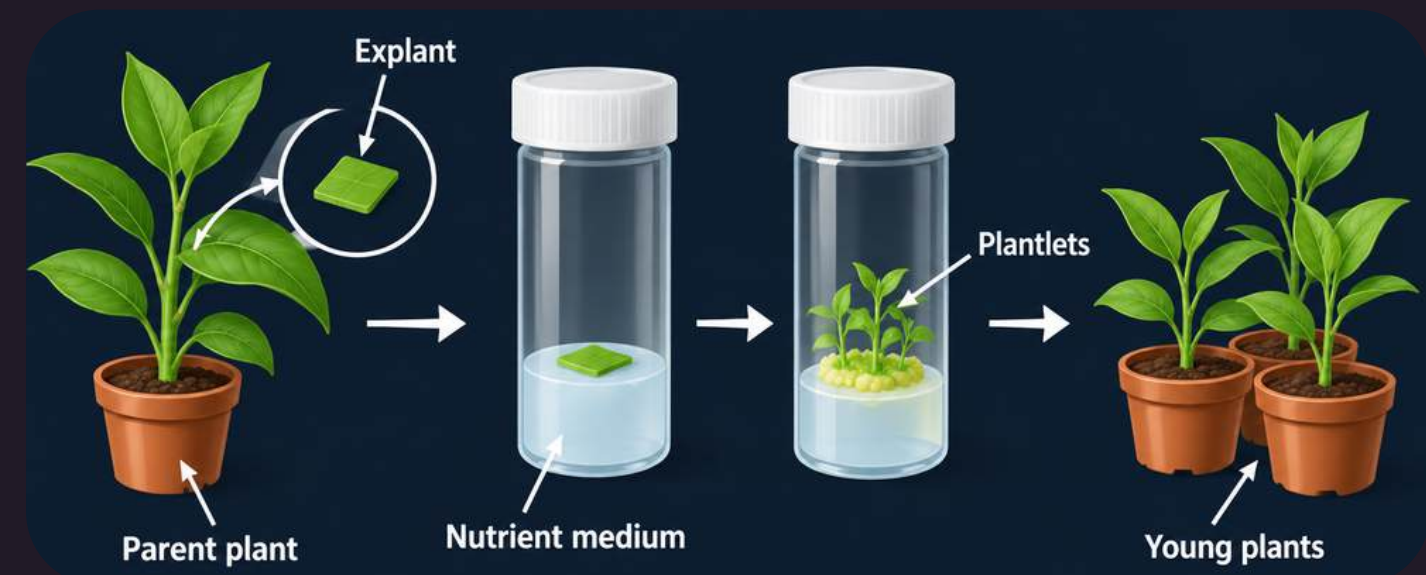
Layering:

- A branch of a plant is bent to the ground and covered with moist soil.
- Roots develop and the buried branch grows into a new plant.
- Example: Jasmine, Rose, Grapes.



Tissue Culture:

- A few cells or tissues are grown in a nutrient medium under laboratory conditions to produce new plants.
- Useful for producing a large number of plants quickly.
- Example: Ornamental plants.



Advantages of Vegetative Propagation

- It is a **quick, easy, and cheap** method of producing a large number of plants in a short time.
- **Genetically identical plants (clones)** to the parent can be produced.
- Plants produced by this method have a **high survival rate**.
- **Desired characteristics of plants** (such as colour, taste, resistance, fragrance, etc.) can be preserved and passed on.
- It helps in the production of **disease-free plants** (especially through techniques like micropropagation/tissue culture).
- Some plants that do not produce viable seeds or take a **long time to mature** can be propagated easily by this method.

Disadvantages of Vegetative Propagation

- No genetic variation in offspring.
- Diseases of parent get transmitted to new plants.
- Not possible in all plants, only certain species.

Check yourself

Ques: Assertion: Vegetative propagation is useful for producing plants that retain the characteristics of the parent.

Reason: Plants produced through this method are genetically similar enough to the parent to retain its characteristics.

Check yourself

Q. Why is vegetative propagation practised for growing some types of plants? [NCERT]

Ans. Vegetative propagation is practised because:

- It helps grow plants that do not produce viable seeds.
- New plants are genetically similar to the parent plant, so desirable characteristics are preserved.
- Such plants generally grow, flower and bear fruits earlier than plants grown from seeds.

Check yourself

Q. A gardener plants seeds collected from a healthy mango tree. After a few years, all the new plants resemble the parent tree, but none of them are completely identical. Which statement best explains this observation?

- (a) The offspring inherit no characteristics from the parent.**
- (b) Reproduction produces offspring that are similar to, but not exact copies of, their parents.**
- (c) All plants produced from seeds are genetically identical.**
- (d) All similarities between the parent and offspring are due only to environmental effects.**

Check yourself

2016, 2017, 2019, 2020, 2023, 2024, 2025

Q. The plants that can be raised by the method of vegetative propagation are:

- (a) Sugarcane, roses, grapes
- (b) Sugarcane, mustard, potato
- (c) Banana, orange, mustard
- (d) Papaya, mustard, potato

Check yourself

- Q. (i) Sugarcane does not produce seeds so name the process through which it will be able to reproduce.
- (ii) List any two advantages of this method.
- (iii) Besides sugarcane, give two more examples of plants that reproduce by this method.

(i) Vegetative propagation.

- (ii) Advantages:** Plants raised by vegetative propagation can bear flowers and fruits earlier than those produced from seeds. Plants produced by vegetative propagation are genetically similar to the parent plant. Plants that have lost the capacity to produce seed can reproduce by vegetative propagation. (any two)
- (iii) Roses, grapes, banana, orange, jasmine etc. (any two examples)**

Check yourself

Q. A plant breeder wants to rapidly multiply a plant that has lost its ability to produce seeds. The new plants should retain the characteristics of the parent plant and may produce flowers and fruits earlier than plants grown from seeds.

- (a) Which method of reproduction should the breeder use?**
- (b) Name any two vegetative parts of a plant that can be used in this method.**
- (c) State one additional advantage of this method.**

(a) Vegetative propagation.

(b) Stem and root. (Leaf can also be used.)

(c) It helps produce a large number of genetically similar plants in a short time.

Check yourself

2015, 2016, 2017, 2019, 2020, 2022, 2024, 2025

Q.(i) What happens when:

(1) Leaves of Bryophyllum fall on the soil?

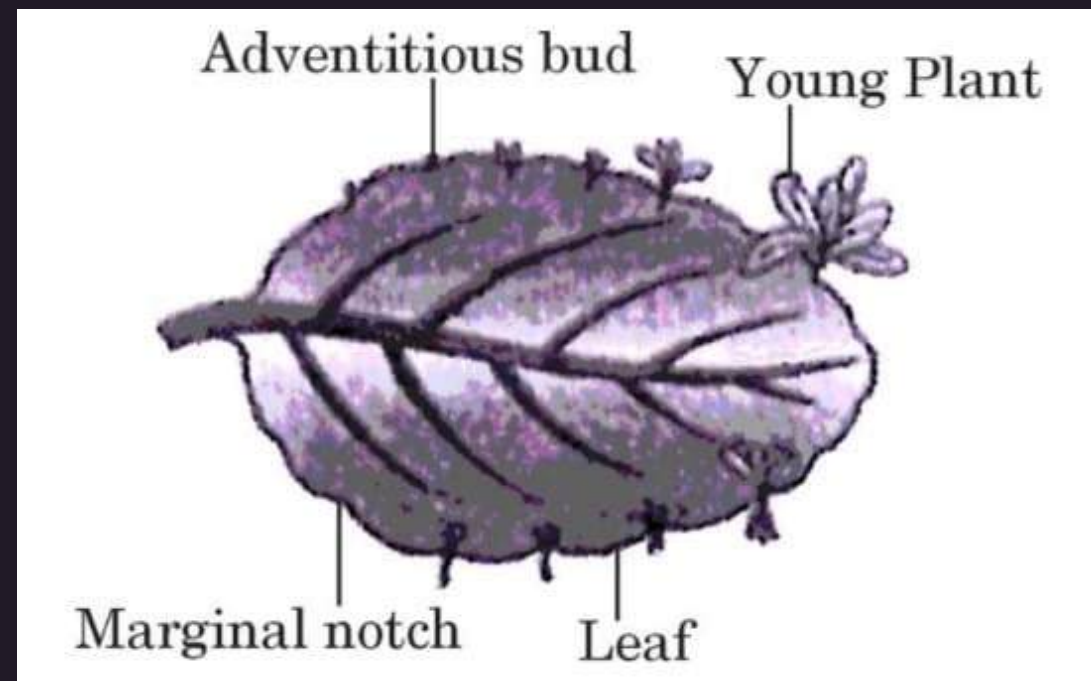
(2) Planaria is cut into many pieces.

(3) Sporangia of Rhizopus on maturation liberate spores? Mention the modes of reproduction in each of the above three cases.

Case	What Happens	Mode of Reproduction
(1) <i>Leaves of Bryophyllum</i> fall on the soil	Buds present along the leaf margins grow into new plants.	Vegetative Propagation (by leaves)
(2) <i>Planaria</i> is cut into many pieces	Each piece regenerates into a complete organism.	Regeneration
(3) <i>Sporangia of Rhizopus</i> liberate spores	Spores germinate under favourable conditions to form new fungal individual	Spore Formation

Check yourself

Q. (a). Observe the given diagram and answer the questions that follow:[5 Marks]



- (i) Give the name of the structure and type of reproduction shown in the above diagram.
- (ii) State one advantage and one disadvantage of the above mentioned method of reproduction.
- (iii) Give reason: Asexual reproduction lacks genetic variation.
- (iv) Why have higher organisms resorted to sexual reproduction in spite of its complexity?

Mode of Asexual Reproduction	Example(s)	Main Features / Description
Binary Fission	<i>Amoeba, Leishmania</i>	Parent cell divides into two equal halves . In <i>Amoeba</i> division can occur in any plane ; in <i>Leishmania</i> it occurs in a definite orientation along the flagellum.
Multiple Fission	<i>Plasmodium</i>	Parent cell divides into many daughter cells simultaneously, usually inside a cyst during unfavourable conditions.
Fragmentation	<i>Spirogyra</i>	The body breaks into fragments on maturation; each fragment grows into a new individual under favourable conditions.
Regeneration	<i>Hydra, Planaria</i>	Cut or broken body parts can grow into complete organisms using specialised cells that multiply and differentiate.
Budding	<i>Hydra, Yeast</i>	A small bud develops on the parent body, grows, and detaches to form a new individual.
Spore Formation	<i>Rhizopus</i> (bread mould)	Sporangia form spores with thick protective walls ; on release and in moisture, spores germinate into new organisms .
Vegetative Propagation	<i>Potato, Bryophyllum, Sweet Potato</i>	New plants arise from vegetative parts (stem, leaf, root) of the parent plant; helps preserve desirable traits.



Bye Bye!